Appendix C. Cumulative impacts of mining and wind power development on reindeer: habitat selection and movement at patch scale (integrated step selection analysis, iSSA)

Here we provide supplementary information regarding the assessment of reindeer habitat selection for grazing sites and movement patterns in the Malå reindeer herding community, Sweden, under the presence of mining and wind power development, for the three snow-free seasons: calving, summer, and autumn. The analysis consisted of an integrated step-selection analysis [iSSA; Avgar et al. (2016)] to simultaneously quantify effects on habitat selection at the foraging patch scale and on movement parameters. We present, for each season: (i) the reindeer GPS data used for the analyses; (ii) the setup of the iSSA models; (iii) the best-ranged iSSA models; (iv) the model selection tables; (v) the tables of coefficients; (vi) figures illustrating the impacts of mining and wind turbines.

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1 Table C1: GPS data used in the iSSA analysis

Year	Phase	Calving	Summer	Autumn
2008	Before construction	47 (17360)	42 (25157)	39 (23666)
2009	Before construction	34 (14370)	33 (16469)	39 (13947)
2010	Construction	40 (17416)	43 (20416)	27 (9833)
2011	Construction	14 (5847)	12 (5143)	9 (4037)
2015	Operation	40 (22767)	38 (19777)	37 (19011)
2016	Operation	31 (11576)	19 (6936)	12 (2825)
2017	Operation	5 (2431)	7 (4092)	11 (4630)
2018	Operation	7 (3180)	6 (3023)	2 (1134)

Table C1. Number of reindeer monitored (number of GPS positions recorded, in parenthesis) in each year and season, used for the integrated step selection analysis. The numbers are sometimes lower than those used in the regional-scale habitat selection analysis.

2 Text C1: Statistical modeling of the iSSA

As described in the main text, the patch-scale habitat selection analysis followed a iSSA. The overall model structure we fitted was the following:

```r
base_formula <- case_ ~ strata(step_id_) +
  # movement
  sl_ + log_sl + cos_ta +
  wind_turbines_YYY_start_construction_scaled:(sl_ + log_sl + cos_ta) +
  wind_turbines_ZZZ_start_operation_scaled:(sl_ + log_sl + cos_ta) +
  landcover_start:(sl_ + log_sl + cos_ta) +
  # habitat selection
  dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled +
  sos_end_scaled +
  log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end +
  log_house_dist_end +
  landcover_end +
  # minig habitat selection
  mining_Krist_nearestXXX_end_scaled +
  # wind habitat selection
  wf_name_end +
  wind_turbines_YYY_end_construction_scaled +
  wind_turbines_ZZZ_end_operation_scaled +
  landcover_end:wind_turbines_YYY_end_construction_scaled +
  landcover_end:wind_turbines_ZZZ_end_operation_scaled +
  # tpi habitat selection
  tpi_scale_TTT +
  # barriers to movement (along the steps)
  public_road_nearest100_cross +
  wind_turbines_nearest500_cross_construction +
  wind_turbines_nearest500_cross_operation
```

where:

- the terms `start`, `end`, and `cross` refer to variables extracted at the start or end points of a step, or along a step. Variables representing habitat selection were selected at the end points of steps, and variables in interaction with movement parameters were extracted at the starting point of steps. Variables with `cross` represented whether an infrastructure was crossed or not along a step and are used to estimate the barrier effect of that infrastructure type;
- terms with `scaled` were scaled to mean = 0 and standard deviation = 1;
- `case_`: our response variable (0/1);
- `dem`: elevation, we include a linear and a quadratic term;
- `slope`: slope;
- `cos_aspect`: cosine of the aspect;
- `tpi_scale_TTT`: topographic position index (TPI) computed at a certain scale, classified in 6 levels: flat (reference level), lower slope, medium slope, upper slope, ridge, valley;

- `sos`: start of season (used for calving and summer), replaced by `eos`, end of season, for autumn;
- `wf_name`: name of the wind power plant closest to each point, with four categories, added as a fixed effect to control for differential effects close to the Åmliden wind park, which lies along a large fenced area;
- `landcover`: land cover, with 5 categories: deciduous forest (reference category), coniferous forest, open areas, mires, and anthropogenic areas;
- `log_public_road_dist`, `log_priv_road_dist`, `log_house_dist`, `log_power_lines_dist`: logarithm of the distance to the nearest infrastructure for public roads, private roads, houses, and power lines;
- `mining_Krist_nearestXXX`: ZOI (exponential decay distance) of the mine, computed considering a certain ZOI radius;
- `sl_`, `log_sl` and `cos_ta`: movement parameters (step length, log(step length), and cosine of the turning angle, respectively).
- `wind_turbines_YYY_end_construction` and `wind_turbines_YYY_end_operation`: variables representing the ZOI of wind turbines. These variables assume a value equal to the wind turbine influence (density or nearest, with a corresponding ZOI radius) at the construction and operation phases, respectively, and were zero for other development phases;
- `public_road_nearest100_cross`: binary variable (0/1) representing whether a certain step crossed or not a public road;
- `wind_turbines_nearest500_cross_construction` and `wind_turbines_nearest500_cross_operation`: binary variables (0/1) representing whether a certain step crossed or not a buffer of 500 m around any wind turbines, during construction and operation.

The model structure contains the movement parameters, including its interaction with the effects of land cover and wind turbines; habitat selection considering the effects of wind turbines in each phase (construction, operation) in interaction with land cover; and main effects of other infrastructure and landscape variables.

The terms `XXX`, `YYY`, `ZZZ`, and `TTT` in the model structure above represent scales or variables that varied across models fitted. To evaluate the most likely ZOI for the mine and the wind turbines, whether there was a cumulative impact of multiple wind power plants, and at which scale the TPI most affected reindeer habitat selection and movement, we replaced these variables in the model structure above using each possible combination of ZOI radii and type (cumulative vs. nearest) for wind power and the different ZOI radii for mining, together with different scales for the TPI (see Fig. B1 in Appendix B). Models were compared through Akaike information criterion (AIC), as described in the main text. The scales included were:

- Mining (`XXX`): 100 m, 250 m, 500 m, 1 km, 2.5 km, 5 km, 10 km.
- Wind turbines (`ZZZ`): density of turbines or distance to the nearest turbine, with ZOI radii 100 m, 250 m, 500 m, 1 km, 2.5 km, 5 km, 10 km.
- TPI: 150 m, 250 m, 510 m.

3 Text C2: Best-ranked integrated step-selection models

This is the structure of the best-ranked iSSA model selected for each season.

Summary of best ranked models:

- **Calving. Cumulative impact of wind turbines, ZOI radius = 5 km; no significant effect of the mine:**
  case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_s
  + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_
  + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) +
  dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled +
  log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end +
  log_house_dist_end + mining_Krist_nearest250_end_scaled + wf_name_end + land-
  cover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled +
  wind_turbines_cumulative5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cum
  + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_operation_scaled
  + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction +
  wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
- **Summer. ZOI of the nearest wind turbine, radius = 5 km; ZOI of the mine with radius 10 km:**
  case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scale
  + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ +
  log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) +
  dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled +
  log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end
  + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end +
  landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled +
  wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest
  + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_scaled +
  public_road_nearest100_cross + wind_turbines_nearest500_cross_construction +
  wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
- **Autumn. Cumulative ZOI of wind turbines, radius = 10km; no significant effect of the mine:**
  case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + dem_end_scaled +
  I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end
  + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end
  + log_mining_Krist_dist_end + wf_name_end + landcover_cc_grouped_end +
  wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_opera
  + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction +
  wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end + sl_:wind_turbines_cumulative10000
  + log_sl:wind_turbines_cumulative10000_start_construction_scaled + cos_ta:wind_turbines_cumulative

+ sl_:wind_turbines_cumulative10000_start_operation_scaled + log_sl:wind_turbines_cumulative10000_
+ cos_ta:wind_turbines_cumulative10000_start_operation_scaled + sl_:landcover_cc_grouped_start
+ log_sl:landcover_cc_grouped_start + cos_ta:landcover_cc_grouped_start + land-
cover_cc_grouped_end:wind_turbines_cumulative10000_end_construction_scaled +
landcover_cc_grouped_end:wind_turbines_cumulative10000_end_operation_scaled

4 Table C2: Model selection table in calving

Table C2 was truncated to show only the first 10 models, to avoid an overly long table.

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
250	cumulative5000	150m_end	442154.7	0.00	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest250_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + land-cover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
10000	cumulative5000	150m_end	442155.0	0.30	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + land-cover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
500	cumulative5000	150m_end	442156.4	1.68	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + land-cover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
1000	cumulative5000	150m_end	442158.2	3.51	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest1000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
2500	cumulative5000	150m_end	442158.3	3.61	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest2500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
5000	cumulative5000	150m_end	442158.3	3.67	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
100	cumulative5000	150m_end	442166.6	11.91	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest100_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative5000_end_construction_scaled + wind_turbines_cumulative5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_construct + landcover_cc_grouped_end:wind_turbines_cumulative5000_end_operation + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
250	nearest5000	150m_end	442174.1	19.42	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest250_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
10000	nearest5000	150m_end	442175.3	20.60	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
500	nearest5000	150m_end	442175.9	21.22	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

Table C2. Table of model selection for the iSSA models for calving. Columns are zone of influence (ZOI) radius for mining and wind turbines, scale for the topographic position index (TPI), AIC of the model, dAIC of each model in relation to the most parsimonious model, and the formula with each model's structure. For impact of wind turbines, cumulative represents the density of turbines, and nearest represents the distance-decay ZOI of the nearest turbine.

5 Table C3: Model selection table in summer

Table C3 was truncated to show only the first 10 models, to avoid an overly long table.

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
10000	nearest5000	150m	470896.2	0.00	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
10000	nearest10000	150m	470896.8	0.63	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest10000_end_construction_scaled + wind_turbines_nearest10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest10000_end_construction + landcover_cc_grouped_end:wind_turbines_nearest10000_end_operation_s + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
5000	nearest5000	150m	470897.9	1.69	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
5000	nearest10000	150m	470898.9	2.70	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest10000_end_construction_scaled + wind_turbines_nearest10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest10000_end_construction + landcover_cc_grouped_end:wind_turbines_nearest10000_end_operation_s + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
2500	nearest5000	150m	470902.8	6.55	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest2500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
10000	nearest2500	150m	470903.3	7.09	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest2500_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest2500_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest2500_end_construction_scaled + wind_turbines_nearest2500_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest2500_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest2500_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
2500	nearest10000	150m	470904.2	7.99	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest2500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest10000_end_construction_scaled + wind_turbines_nearest10000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_nearest10000_end_construction + land-cover_cc_grouped_end:wind_turbines_nearest10000_end_operation_s + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
5000	nearest2500	150m	470904.9	8.68	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest2500_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest2500_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest2500_end_construction_scaled + wind_turbines_nearest2500_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_nearest2500_end_construction_ + land-cover_cc_grouped_end:wind_turbines_nearest2500_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
100	nearest5000	150m	470905.6	9.36	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest5000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest5000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest100_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest5000_end_construction_scaled + wind_turbines_nearest5000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest5000_end_construction_ + landcover_cc_grouped_end:wind_turbines_nearest5000_end_operation_sc + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end
10000	cumulative10000	150m	470906.2	10.03	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + sos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s150m_end

Table C3. Table of model selection for the iSSA models for summer. Columns are zone of influence (ZOI) radius for mining and wind turbines, scale for the topographic position index (TPI), AIC of the model, dAIC of each model in relation to the most parsimonious model, and the formula with each model's structure. For impact of wind turbines, cumulative represents the density of turbines, and nearest represents the distance-decay ZOI of the nearest turbine. Here, we selected model ranked number 3 as the most parsimonious, because delta AIC < 2 in relation to the best ranked model and we chose to be more conservative, i.e. to select the smallest ZOI radii.

6 Table C4: Model selection table in autumn

Table C4 was truncated to show only the first 10 models, to avoid an overly long table.

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
10000	cumulative10000	510m	360554.8	0.00	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end
5000	cumulative10000	510m	360555.4	0.55	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
100	cumulative10000	510m	360555.5	0.60	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest100_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + land-cover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end
250	cumulative10000	510m	360555.6	0.71	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest250_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + land-cover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
2500	cumulative10000	510m	360555.8	0.93	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest2500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end
500	cumulative10000	510m	360556.1	1.30	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest500_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + landcover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end

(continued)

Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
1000	cumulative10000	510m	360556.8	1.94	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_cumulative10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_cumulative10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest1000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_cumulative10000_end_construction_scaled + wind_turbines_cumulative10000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_cumulative10000_end_construc + land-cover_cc_grouped_end:wind_turbines_cumulative10000_end_operatio + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end
10000	nearest10000	510m	360564.5	9.68	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest10000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest10000_end_construction_scaled + wind_turbines_nearest10000_end_operation_scaled + land-cover_cc_grouped_end:wind_turbines_nearest10000_end_construction + land-cover_cc_grouped_end:wind_turbines_nearest10000_end_operation_s + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end

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Mining ZOI radius	Wind turbine ZOI radius	TPI	AIC	dAIC	formula
5000	nearest10000	510m	360565.1	10.27	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest5000_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest10000_end_construction_scaled + wind_turbines_nearest10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest10000_end_construction + landcover_cc_grouped_end:wind_turbines_nearest10000_end_operation_s + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end
100	nearest10000	510m	360565.3	10.50	case_ ~ strata(step_id_) + sl_ + log_sl + cos_ta + wind_turbines_nearest10000_start_construction_scaled:(sl_ + log_sl + cos_ta) + wind_turbines_nearest10000_start_operation_scaled:(sl_ + log_sl + cos_ta) + landcover_cc_grouped_start:(sl_ + log_sl + cos_ta) + dem_end_scaled + I(dem_end_scaled^2) + slope_end_scaled + eos_end_scaled + log_power_lines_dist_end + log_priv_road_dist_end + log_public_road_dist_end + log_house_dist_end + mining_Krist_nearest100_end_scaled + wf_name_end + landcover_cc_grouped_end + wind_turbines_nearest10000_end_construction_scaled + wind_turbines_nearest10000_end_operation_scaled + landcover_cc_grouped_end:wind_turbines_nearest10000_end_construction + landcover_cc_grouped_end:wind_turbines_nearest10000_end_operation_s + public_road_nearest100_cross + wind_turbines_nearest500_cross_construction + wind_turbines_nearest500_cross_operation + tpi_cat_s510m_end

Table C4. Table of model selection for the iSSA models for autumn. Columns are zone of influence (ZOI) radius for mining and wind turbines, scale for the topographic position index (TPI), AIC of the model, dAIC of each model in relation to the most parsimonious model, and the formula with each model's structure. For impact of wind turbines, cumulative represents the density of turbines, and nearest represents the distance-decay ZOI of the nearest turbine.

7 Text C3: Disturbance effects on step selection

Here we present the coefficients of the iSSA models on habitat selection at patch scale. Even though all results come from the same models, we split the way we present the parts of the resulting model, to ease its interpretation. In this section we restrict ourselves on presenting the coefficients of all infrastructure and landscape covariates, without going into details regarding the impact of wind turbines on habitat selection and on movement patterns, which are shown further down.

Table C5 shows the coefficient estimates for all the infrastructure types, excluding wind turbines. Wind turbines are only included here for the barrier effects. The estimates are put together for the different seasons in table C5 so it is easy to compare the effects between them.

A summary of the results:

- Wind power plants were barriers to reindeer movement in calving during construction (and marginally during operation; p = 0.07) and in summer during wind turbine operation;
- Public roads were barrier in all seasons;
- The mine was avoided in summer (even though very weakly, Fig. C2), but not in calving and autumn;
- Public and private roads were avoided in all seasons;
- There was no effect of power lines in calving and summer, and a slight selection for them during autumn;
- Houses were selected during calving and autumn; (*why??*)
- Start of season did not affect habitat selection during calving and summer; however, end of season was positively selected in autumn, which means animals select for areas which stay green longer in autumn.

season	term	estimate	std.error	p	select
autumn	eos_end_scaled	0.097	0.010	<0.001	selection/positive
calving	log_house_dist_end	-0.213	0.008	<0.001	selection/positive
summer	log_house_dist_end	0.010	0.009	0.279	no effect
autumn	log_house_dist_end	-0.051	0.010	<0.001	selection/positive
calving	log_power_lines_dist_end	0.005	0.012	0.705	no effect
summer	log_power_lines_dist_end	0.002	0.007	0.734	no effect
autumn	log_power_lines_dist_end	-0.023	0.008	0.004	selection/positive
calving	log_priv_road_dist_end	0.030	0.004	<0.001	avoidance/negative
summer	log_priv_road_dist_end	0.043	0.004	<0.001	avoidance/negative
autumn	log_priv_road_dist_end	0.012	0.004	0.008	avoidance/negative
calving	log_public_road_dist_end	0.066	0.007	<0.001	avoidance/negative
summer	log_public_road_dist_end	0.051	0.008	<0.001	avoidance/negative
autumn	log_public_road_dist_end	0.041	0.009	<0.001	avoidance/negative
autumn	mining_Krist_nearest10000_end_scaled	-0.042	0.030	0.162	no effect
calving	mining_Krist_nearest250_end_scaled	-1171.291	890.424	0.188	no effect
summer	mining_Krist_nearest5000_end_scaled	-0.044	0.013	0.001	avoidance/negative
calving	public_road_nearest100_cross	-0.118	0.022	<0.001	avoidance/negative
summer	public_road_nearest100_cross	-0.150	0.022	<0.001	avoidance/negative

(continued)

season	term	estimate	std.error	p	select
autumn	public_road_nearest100_cross	-0.140	0.024	<0.001	avoidance/negative
calving	sos_end_scaled	-0.001	0.009	0.922	no effect
summer	sos_end_scaled	-0.001	0.007	0.931	no effect
calving	wind_turbines_nearest500_cross_construction	-0.593	0.209	0.005	avoidance/negative
summer	wind_turbines_nearest500_cross_construction	-0.007	0.092	0.939	no effect
autumn	wind_turbines_nearest500_cross_construction	0.138	0.170	0.415	no effect
calving	wind_turbines_nearest500_cross_operation	-0.221	0.122	0.07	no effect
summer	wind_turbines_nearest500_cross_operation	-0.173	0.066	0.009	avoidance/negative
autumn	wind_turbines_nearest500_cross_operation	-0.018	0.116	0.877	no effect

Table C5. Coefficient estimates for the effects of infrastructure types and the start/end of season on habitat selection for all three seasons.

8 Table C6: Coefficients - calving

term	estimate	std.error	p	select
sl_	0.000	0.000	<0.001	no effect
log_sl	0.027	0.002	<0.001	avoidance/negative
cos_ta	-0.053	0.005	<0.001	avoidance/negative
dem_end_scaled	-0.073	0.014	<0.001	avoidance/negative
I(dem_end_scaled^2)	-0.020	0.005	<0.001	avoidance/negative
slope_end_scaled	-0.054	0.005	<0.001	avoidance/negative
sos_end_scaled	-0.001	0.009	0.922	no effect
log_power_lines_dist_end	0.005	0.012	0.705	no effect
log_priv_road_dist_end	0.030	0.004	<0.001	avoidance/negative
log_public_road_dist_end	0.066	0.007	<0.001	avoidance/negative
log_house_dist_end	-0.213	0.008	<0.001	selection/positive
mining_Krist_nearest250_end_scaled	-1171.291	890.424	0.188	no effect
wf_name_endstorliden	-0.288	0.036	<0.001	avoidance/negative
wf_name_endytteberg	-0.026	0.051	0.61	no effect
wf_name_endamliden	-0.293	0.083	<0.001	avoidance/negative
landcover_cc_grouped_endopen lands	0.621	0.009	<0.001	selection/positive
landcover_cc_grouped_endanthropogenic areas	0.313	0.028	<0.001	selection/positive
landcover_cc_grouped_endconiferous forests	-0.105	0.008	<0.001	avoidance/negative
landcover_cc_grouped_endmires	-0.051	0.009	<0.001	avoidance/negative
wind_turbines_cumulative5000_end_construction_scaled	-0.025	0.019	0.178	no effect
wind_turbines_cumulative5000_end_operation_scaled	0.017	0.020	0.413	no effect
public_road_nearest100_cross	-0.118	0.022	<0.001	avoidance/negative
wind_turbines_nearest500_cross_construction	-0.593	0.209	0.005	avoidance/negative
wind_turbines_nearest500_cross_operation	-0.221	0.122	0.07	no effect
tpi_cat_s150m_endlower slope	-0.042	0.012	<0.001	avoidance/negative
tpi_cat_s150m_endmedium slope	0.105	0.008	<0.001	selection/positive
tpi_cat_s150m_endridge	0.131	0.012	<0.001	selection/positive
tpi_cat_s150m_endupper slope	0.114	0.012	<0.001	selection/positive

(continued)

term	estimate	std.error	p	select
tpi_cat_s150m_endvalley	-0.022	0.015	0.138	no effect
sl_:wind_turbines_cumulative5000_start_construction_scaled	0.000	0.000	0.005	no effect
log_sl:wind_turbines_cumulative5000_start_construction_scaled	0.019	0.003	<0.001	avoidance/negative
cos_ta:wind_turbines_cumulative5000_start_construction_scaled	0.011	0.005	0.033	selection/positive
sl_:wind_turbines_cumulative5000_start_operation_scaled	0.000	0.000	0.051	no effect
log_sl:wind_turbines_cumulative5000_start_operation_scaled	0.006	0.003	0.02	avoidance/negative
cos_ta:wind_turbines_cumulative5000_start_operation_scaled	-0.003	0.005	0.585	no effect
sl_:landcover_cc_grouped_startopen lands	0.000	0.000	<0.001	no effect
sl_:landcover_cc_grouped_startanthropogenic areas	0.001	0.000	<0.001	selection/positive
sl_:landcover_cc_grouped_startconiferous forests	0.000	0.000	0.194	no effect
sl_:landcover_cc_grouped_startmires	0.000	0.000	0.038	no effect
log_sl:landcover_cc_grouped_startopen lands	0.081	0.004	<0.001	avoidance/negative
log_sl:landcover_cc_grouped_startanthropogenic areas	-0.006	0.014	0.687	no effect
log_sl:landcover_cc_grouped_startconiferous forests	-0.094	0.004	<0.001	selection/positive
log_sl:landcover_cc_grouped_startmires	0.002	0.006	0.71	no effect
cos_ta:landcover_cc_grouped_startopen lands	0.012	0.009	0.147	no effect
cos_ta:landcover_cc_grouped_startanthropogenic areas	-0.494	0.032	<0.001	avoidance/negative
cos_ta:landcover_cc_grouped_startconiferous forests	-0.006	0.008	0.5	no effect
cos_ta:landcover_cc_grouped_startmires	0.079	0.011	<0.001	selection/positive
landcover_cc_grouped_endopen lands:wind_turbines_cumulative5000_end_construction_scaled	0.003	0.008	0.718	no effect
landcover_cc_grouped_endanthropogenic areas:wind_turbines_cumulative5000_end_construction_scaled	0.035	0.024	0.146	no effect
landcover_cc_grouped_endconiferous forests:wind_turbines_cumulative5000_end_construction_scaled	-0.036	0.009	<0.001	avoidance/negative
landcover_cc_grouped_endmires:wind_turbines_cumulative5000_end_constru	0.041	0.010	<0.001	selection/positive
landcover_cc_grouped_endopen lands:wind_turbines_cumulative5000_end_operation_scaled	-0.039	0.008	<0.001	avoidance/negative
landcover_cc_grouped_endanthropogenic areas:wind_turbines_cumulative5000_end_operation_scaled	-0.021	0.028	0.451	no effect
landcover_cc_grouped_endconiferous forests:wind_turbines_cumulative5000_end_operation_scaled	-0.033	0.008	<0.001	avoidance/negative
landcover_cc_grouped_endmires:wind_turbines_cumulative5000_end_operati	-0.017	0.009	0.073	no effect

Table C6. Complete table with coefficient estimates for the best-ranked integrated step selection model in calving.

9 Table C7: Coefficients - summer

term	estimate	std.error	p	select
sl_	0.000	0.000	0.99	no effect
log_sl	-0.002	0.012	0.877	no effect
cos_ta	-0.055	0.013	<0.001	avoidance/negative
dem_end_scaled	0.060	0.012	<0.001	selection/positive
I_dem_end_scaled^2	-0.002	0.005	0.648	no effect
slope_end_scaled	-0.087	0.004	<0.001	avoidance/negative

(continued)

term	estimate	std.error	p	select
sos_end_scaled	-0.001	0.007	0.931	no effect
log_power_lines_dist_end	0.002	0.007	0.734	no effect
log_priv_road_dist_end	0.043	0.004	<0.001	avoidance/negative
log_public_road_dist_end	0.051	0.008	<0.001	avoidance/negative
log_house_dist_end	0.010	0.009	0.279	no effect
mining_Krist_nearest5000_end_scaled	-0.044	0.013	0.001	avoidance/negative
wf_name_endstorliden	0.106	0.048	0.028	selection/positive
wf_name_endytteberg	0.078	0.054	0.147	no effect
wf_name_endamliden	0.188	0.065	0.004	selection/positive
landcover_cc_grouped_endopen lands	0.250	0.013	<0.001	selection/positive
landcover_cc_grouped_endanthropogenic areas	0.691	0.027	<0.001	selection/positive
landcover_cc_grouped_endconiferous forests	0.076	0.011	<0.001	selection/positive
landcover_cc_grouped_endmires	-0.499	0.014	<0.001	avoidance/negative
wind_turbines_nearest5000_end_construction_scaled	0.041	0.022	0.06	no effect
wind_turbines_nearest5000_end_operation_scaled	0.073	0.020	<0.001	selection/positive
public_road_nearest100_cross	-0.150	0.022	<0.001	avoidance/negative
wind_turbines_nearest500_cross_construction	-0.007	0.092	0.939	no effect
wind_turbines_nearest500_cross_operation	-0.173	0.066	0.009	avoidance/negative
tpi_cat_s150m_endlower slope	0.083	0.012	<0.001	selection/positive
tpi_cat_s150m_endmedium slope	0.036	0.009	<0.001	selection/positive
tpi_cat_s150m_endridge	0.036	0.013	0.006	selection/positive
tpi_cat_s150m_endupper slope	0.038	0.012	0.002	selection/positive
tpi_cat_s150m_endvalley	0.079	0.014	<0.001	selection/positive
sl_:wind_turbines_nearest5000_start_construction_scaled	0.000	0.000	0.376	no effect
log_sl:wind_turbines_nearest5000_start_construction_scaled	-0.006	0.004	0.212	no effect
cos_ta:wind_turbines_nearest5000_start_construction_scaled	-0.008	0.005	0.095	no effect
sl_:wind_turbines_nearest5000_start_operation_scaled	0.000	0.000	<0.001	no effect
log_sl:wind_turbines_nearest5000_start_operation_scaled	0.021	0.005	<0.001	avoidance/negative
cos_ta:wind_turbines_nearest5000_start_operation_scaled	-0.008	0.005	0.087	no effect
sl_:landcover_cc_grouped_startopen lands	0.000	0.000	0.048	no effect
sl_:landcover_cc_grouped_startanthropogenic areas	0.001	0.000	<0.001	selection/positive
sl_:landcover_cc_grouped_startconiferous forests	0.000	0.000	0.01	no effect
sl_:landcover_cc_grouped_startmires	0.000	0.000	<0.001	no effect
log_sl:landcover_cc_grouped_startopen lands	0.008	0.014	0.573	no effect
log_sl:landcover_cc_grouped_startanthropogenic areas	-0.036	0.026	0.17	no effect
log_sl:landcover_cc_grouped_startconiferous forests	0.080	0.013	<0.001	avoidance/negative
log_sl:landcover_cc_grouped_startmires	-0.132	0.016	<0.001	selection/positive
cos_ta:landcover_cc_grouped_startopen lands	0.051	0.016	0.002	selection/positive
cos_ta:landcover_cc_grouped_startanthropogenic areas	0.001	0.033	0.964	no effect
cos_ta:landcover_cc_grouped_startconiferous forests	0.040	0.015	0.007	selection/positive
cos_ta:landcover_cc_grouped_startmires	0.053	0.019	0.005	selection/positive
landcover_cc_grouped_endopen lands:wind_turbines_nearest5000_end_construction_scaled	0.014	0.013	0.287	no effect
landcover_cc_grouped_endanthropogenic areas:wind_turbines_nearest5000_end_construction_scaled	-0.007	0.023	0.769	no effect
landcover_cc_grouped_endconiferous forests:wind_turbines_nearest5000_end_construction_scaled	0.017	0.012	0.16	no effect
landcover_cc_grouped_endmires:wind_turbines_nearest5000_end_constructi	0.042	0.016	0.008	selection/positive
landcover_cc_grouped_endopen lands:wind_turbines_nearest5000_end_operation_scaled	-0.019	0.011	0.079	no effect

(continued)

term	estimate	std.error	p	select
landcover_cc_grouped_endanthropogenic areas:wind_turbines_nearest5000_end_operation_scaled	-0.010	0.020	0.601	no effect
landcover_cc_grouped_endconiferous forests:wind_turbines_nearest5000_end_operation_scaled	-0.068	0.010	<0.001	avoidance/negative
landcover_cc_grouped_endmires:wind_turbines_nearest5000_end_operation_	-0.061	0.015	<0.001	avoidance/negative

Table C7. Complete table with coefficient estimates for the best-ranked integrated step selection model in summer.

10 Table C8: Coefficients - autumn

term	estimate	std.error	p	select
sl_	0.000	0.000	<0.001	no effect
log_sl	-0.058	0.017	<0.001	selection/positive
cos_ta	0.029	0.023	0.207	no effect
dem_end_scaled	-0.223	0.021	<0.001	avoidance/negative
I(dem_end_scaled^2)	0.077	0.007	<0.001	selection/positive
slope_end_scaled	-0.105	0.005	<0.001	avoidance/negative
eos_end_scaled	0.097	0.010	<0.001	selection/positive
log_power_lines_dist_end	-0.023	0.008	0.004	selection/positive
log_priv_road_dist_end	0.012	0.004	0.008	avoidance/negative
log_public_road_dist_end	0.041	0.009	<0.001	avoidance/negative
log_house_dist_end	-0.051	0.010	<0.001	selection/positive
mining_Krist_nearest10000_end_scaled	-0.042	0.030	0.162	no effect
wf_name_endstorliden	0.199	0.063	0.002	selection/positive
wf_name_endytteberg	0.159	0.069	0.02	selection/positive
wf_name_endamliden	0.541	0.096	<0.001	selection/positive
landcover_cc_grouped_endopen lands	1.009	0.018	<0.001	selection/positive
landcover_cc_grouped_endanthropogenic areas	0.427	0.038	<0.001	selection/positive
landcover_cc_grouped_endconiferous forests	0.482	0.018	<0.001	selection/positive
landcover_cc_grouped_endmires	0.112	0.021	<0.001	selection/positive
wind_turbines_cumulative10000_end_construction_scaled	-0.080	0.048	0.095	no effect
wind_turbines_cumulative10000_end_operation_scaled	0.068	0.044	0.121	no effect
public_road_nearest100_cross	-0.140	0.024	<0.001	avoidance/negative
wind_turbines_nearest500_cross_construction	0.138	0.170	0.415	no effect
wind_turbines_nearest500_cross_operation	-0.018	0.116	0.877	no effect
tpi_cat_s510m_endlower slope	-0.101	0.018	<0.001	avoidance/negative
tpi_cat_s510m_endmedium slope	0.135	0.017	<0.001	selection/positive
tpi_cat_s510m_endridge	0.296	0.015	<0.001	selection/positive
tpi_cat_s510m_endupper slope	0.177	0.018	<0.001	selection/positive
tpi_cat_s510m_endvalley	-0.174	0.016	<0.001	avoidance/negative
sl_:wind_turbines_cumulative10000_start_construction_scaled	0.000	0.000	<0.001	no effect
log_sl:wind_turbines_cumulative10000_start_construction_scaled	-0.011	0.004	0.002	selection/positive
cos_ta:wind_turbines_cumulative10000_start_construction_scaled	0.007	0.006	0.203	no effect
sl_:wind_turbines_cumulative10000_start_operation_scaled	0.000	0.000	0.899	no effect
log_sl:wind_turbines_cumulative10000_start_operation_scaled	0.002	0.004	0.681	no effect

(continued)

term	estimate	std.error	p	select
cos_ta:wind_turbines_cumulative10000_start_operation_scaled	0.004	0.006	0.439	no effect
sl_:landcover_cc_grouped_startopen lands	0.000	0.000	<0.001	no effect
sl_:landcover_cc_grouped_startanthropogenic areas	0.000	0.000	<0.001	no effect
sl_:landcover_cc_grouped_startconiferous forests	0.000	0.000	<0.001	no effect
sl_:landcover_cc_grouped_startmires	0.000	0.000	0.132	no effect
log_sl:landcover_cc_grouped_startopen lands	0.148	0.018	<0.001	avoidance/negative
log_sl:landcover_cc_grouped_startanthropogenic areas	0.138	0.034	<0.001	avoidance/negative
log_sl:landcover_cc_grouped_startconiferous forests	0.124	0.018	<0.001	avoidance/negative
log_sl:landcover_cc_grouped_startmires	-0.157	0.021	<0.001	selection/positive
cos_ta:landcover_cc_grouped_startopen lands	-0.026	0.025	0.3	no effect
cos_ta:landcover_cc_grouped_startanthropogenic areas	-0.102	0.050	0.04	avoidance/negative
cos_ta:landcover_cc_grouped_startconiferous forests	-0.019	0.025	0.436	no effect
cos_ta:landcover_cc_grouped_startmires	-0.019	0.030	0.526	no effect
landcover_cc_grouped_endopen lands:wind_turbines_cumulative10000_end_construction_scaled	0.040	0.019	0.035	selection/positive
landcover_cc_grouped_endanthropogenic areas:wind_turbines_cumulative10000_end_construction_scaled	0.018	0.038	0.637	no effect
landcover_cc_grouped_endconiferous forests:wind_turbines_cumulative10000_end_construction_scaled	0.032	0.019	0.099	no effect
landcover_cc_grouped_endmires:wind_turbines_cumulative10000_end_const	0.026	0.023	0.256	no effect
landcover_cc_grouped_endopen lands:wind_turbines_cumulative10000_end_operation_scaled	0.005	0.016	0.738	no effect
landcover_cc_grouped_endanthropogenic areas:wind_turbines_cumulative10000_end_operation_scaled	-0.057	0.037	0.12	no effect
landcover_cc_grouped_endconiferous forests:wind_turbines_cumulative10000_end_operation_scaled	-0.071	0.016	<0.001	avoidance/negative
landcover_cc_grouped_endmires:wind_turbines_cumulative10000_end_opera	-0.089	0.023	<0.001	avoidance/negative

Table C8. Complete table with coefficient estimates for the best-ranked integrated step selection model in autumn.

11 Figure C1: Impacts of wind power on habitat selection in calving during operation phase

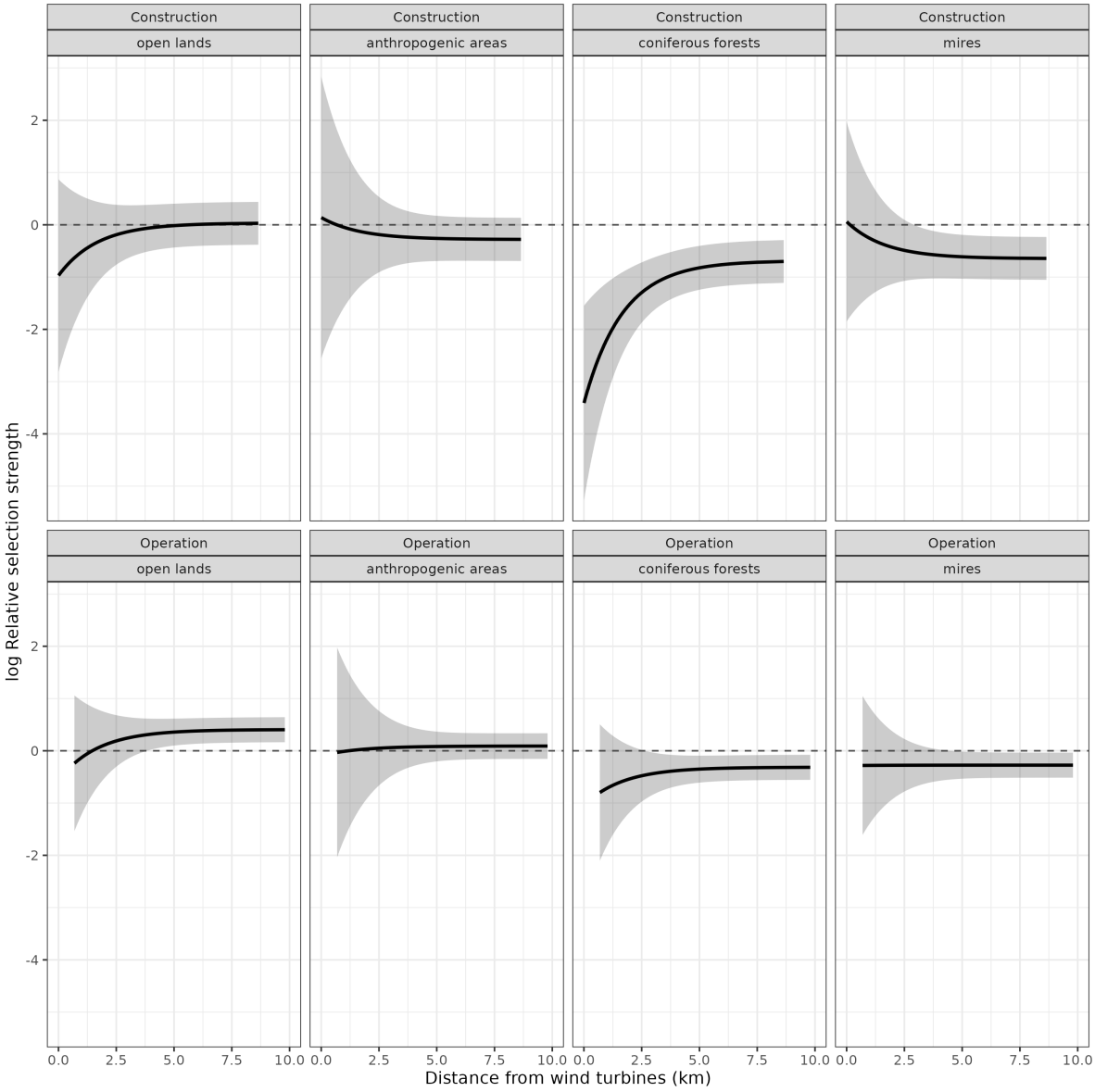


Figure C1. Comparison of response curves for calving during construction and operation of the wind power plants. The curves show the log-Relative selection strength as a function of the distance to wind turbines, considering the impact of 10 wind turbines. The model uses deciduous forests as the reference category.

12 Figure C2: Impacts of mining on habitat selection in summer

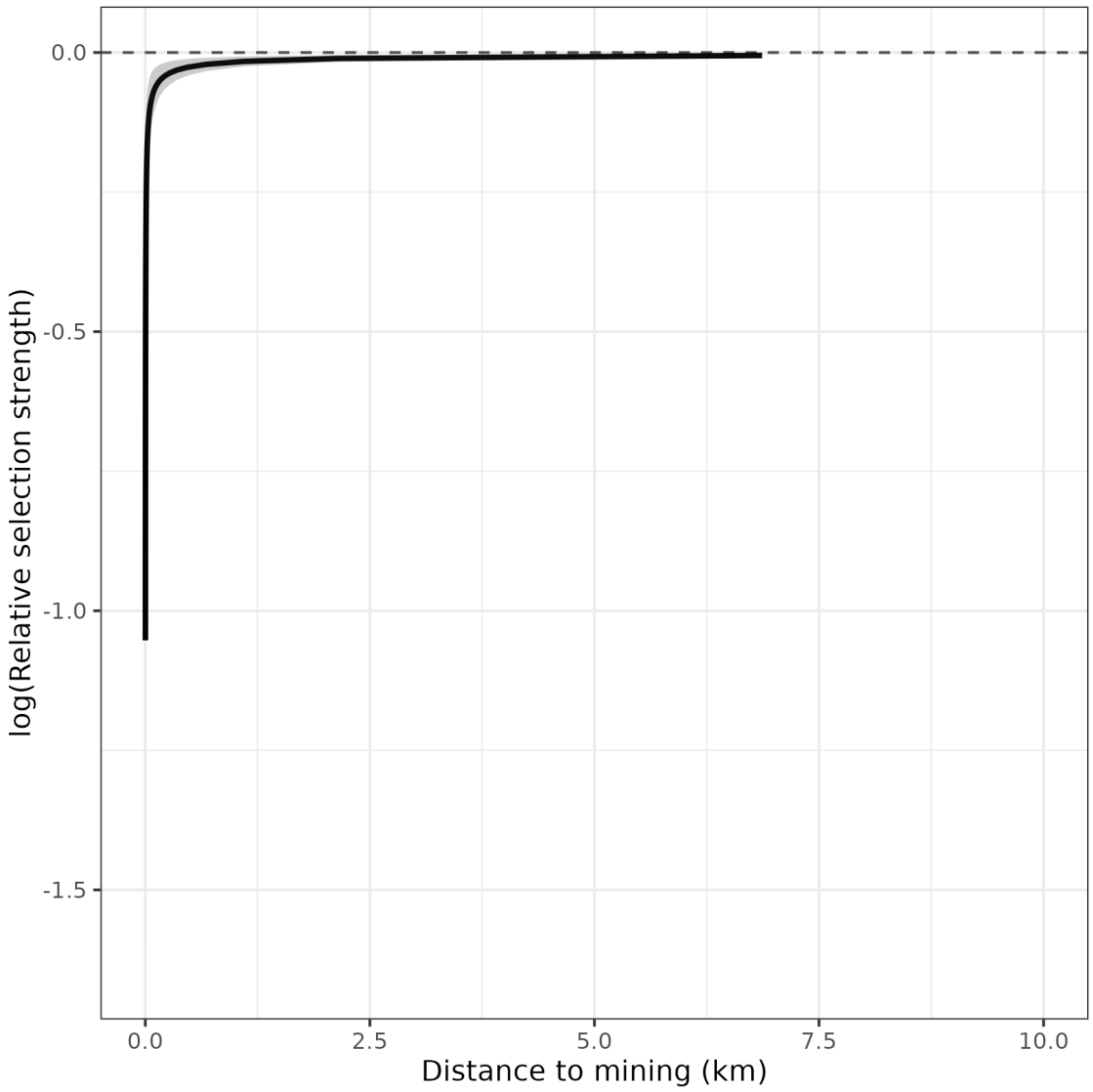


Figure C2. Log-Relative selection strength as a function of the distance to the mine in Kristineberget in summer.

13 Figure C3: Impacts of wind power on habitat selection in summer

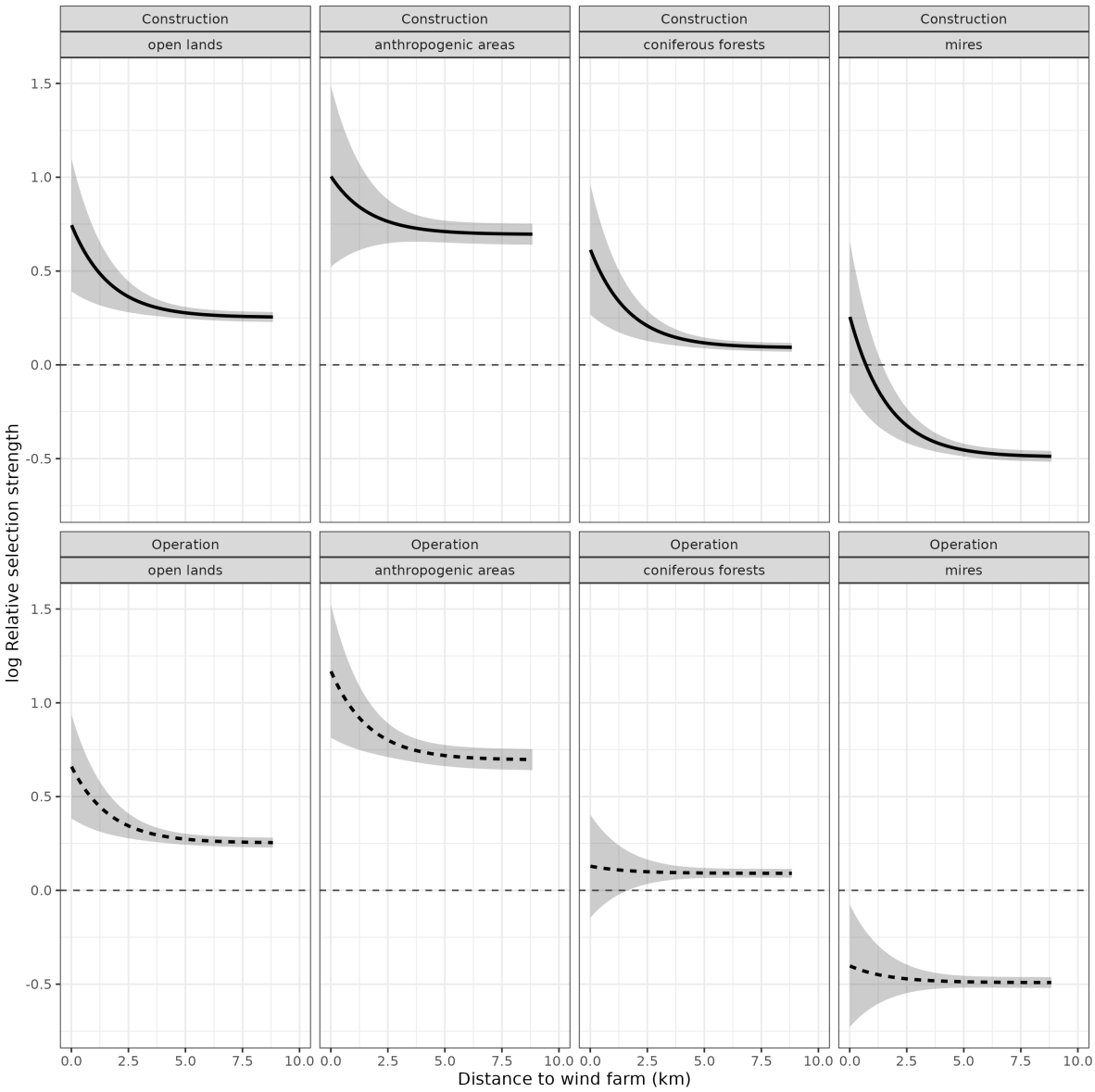


Figure C3. Comparison of response curves for summer during construction and operation of the wind power plants. The curves show the log-Relative selection strength as a function of the distance to the nearest wind turbine. The model uses deciduous forests as the reference category.

14 Figure C4: Impacts of wind power on habitat selection in autumn

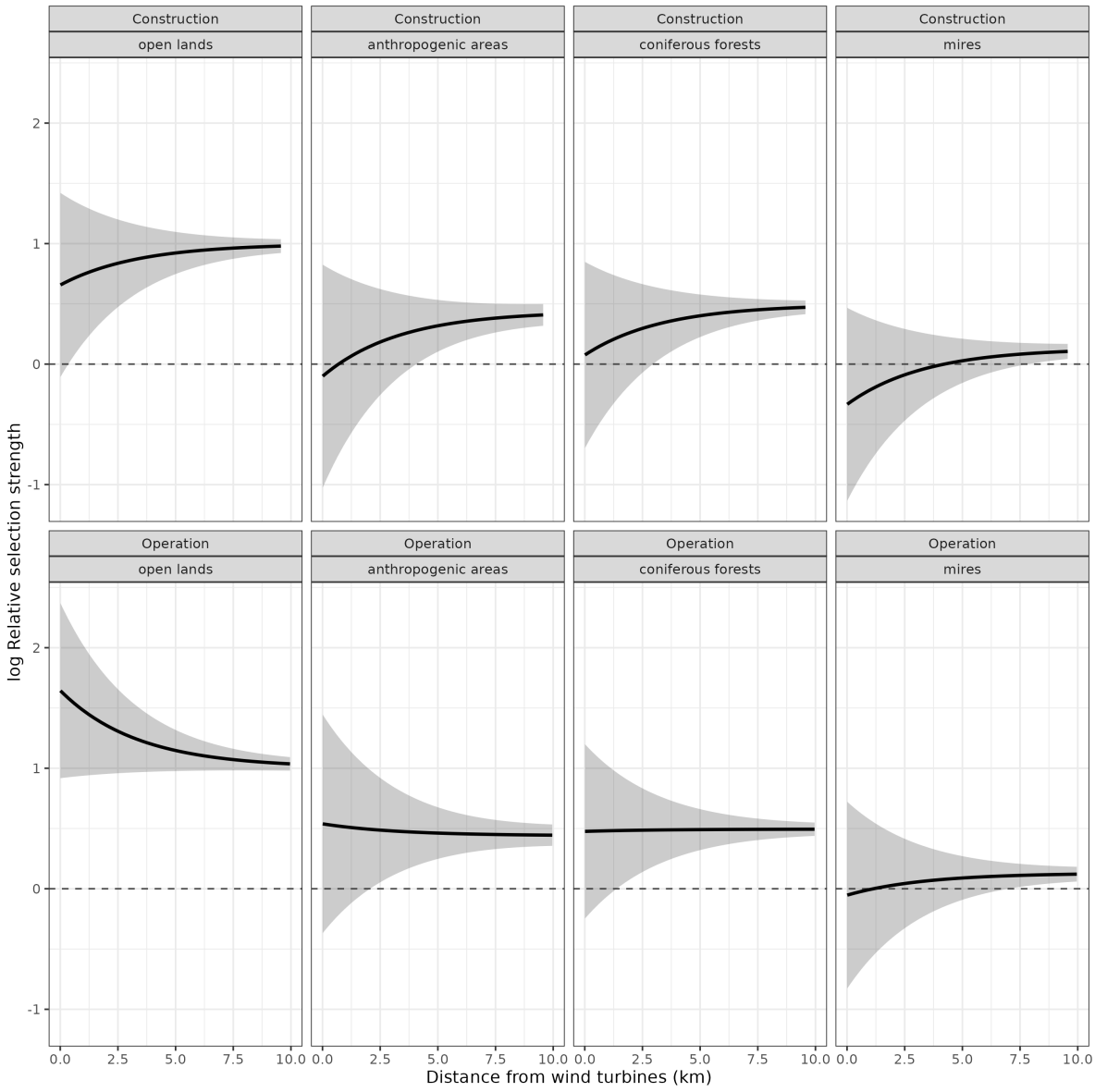


Figure C4. Comparison of response curves for autumn during construction and operation of the wind power plants. The curves show the log-Relative selection strength as a function of the distance to wind turbine, considering the impact of 10 turbines. The model uses deciduous forests as the reference category.

15 Figure C5: Impacts of wind turbines on movement patterns in calving

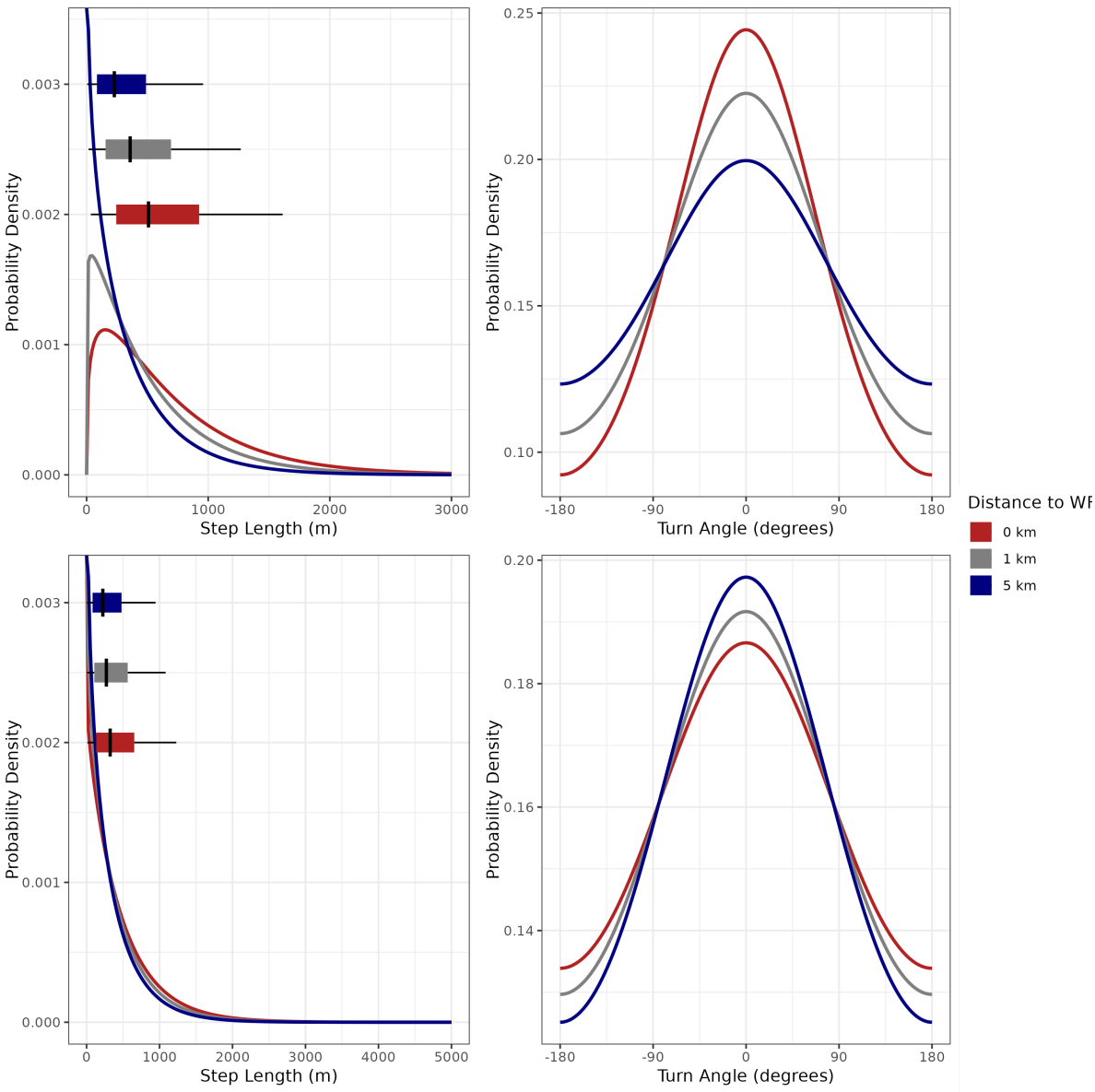


Figure C5. Predicted distribution of step length (left) and turning angles (right) at different distances from where the wind turbines were built, during the construction phase (upper row) and operation phase (bottom row). The plot in the left also shows boxplots of predicted movement rates. Reindeer moved faster and more directionaly when close to wind turbines during their construction, but the different was minimal during operation.

References

Avgar, Tal, Jonathan R. Potts, Mark A. Lewis, and Mark S. Boyce. 2016. "Integrated Step Selection Analysis: Bridging the Gap Between Resource Selection and Animal Movement." Edited by Luca Börger. *Methods in Ecology and Evolution* 7 (5): 619–30. https://doi.org/10.1111/2041-210X.12528.